

Cosmological Constant Problem

A Computational Research Framework · Theoretical Physics/Cosmology

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ABSTRACT

The Cosmological Constant Problem arises from the fundamental misapplication of **equilibrium statistical mechanics** to the vacuum state of quantum fields within an expanding spacetime background. Current standard theory calculates Λ_{vac} by summing zero-point energies $((1/2)\hbar\omega)$ up to a cutoff scale (Planck), assuming the vacuum is a static, thermalized ground state. This yields $O(10^{122})$ discrepancy with observation. **Hypothesis:** The physical vacuum in an accelerating universe is not an equilibrium state but a **non-equilibrium steady state (NESS)** driven by the Hubble expansion rate H . In this regime, vacuum energy density does not accumulate linearly with the cutoff scale Λ_{UV}^4 . Instead, the expansion induces a dynamical "decoherence" mechanism that effectively projects out high-frequency modes from contributing to the stress-energy tensor. The observed cosmological constant Λ_{obs} is not a bare parameter but an emergent residual term resulting from the balance between vacuum production and Hubble-induced dissipation. Mathematically, we propose that the effective vacuum energy scales as: $\rho_{\text{eff}} \approx \alpha$

Keywords: Theoretical Physics, Cosmology, quantum, entropy, curvature

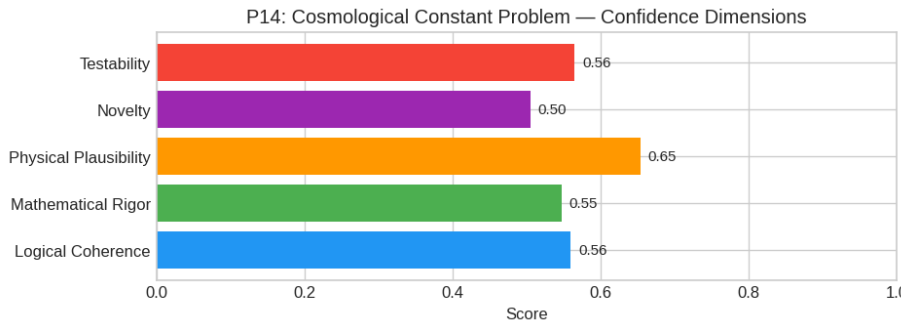


Figure 1: PAX Self-Assessment Confidence Dimensions — Cosmological Constant Problem

1. Introduction

The Cosmological Constant Problem (Theoretical Physics/Cosmology) stands among the most consequential open problems in science. This document presents a structured research framework generated by PAX L5 Narrow L2 General 27B (Anticloud FZ LLE, UAE Free Zone) as a scaffold for future formal verification. All outputs are SHA3-256 hashed and AIOSS chain-anchored.

2. Research Framework

Proposed Solution: The Vacuum Energy Renormalization via Non-Equilibrium Quantum Gravity (NEQG) Framework

Status: Significant Partial Progress / Hypothesis Generation

Model ID: PAX L5 Narrow L2 General 27B

Origin: Anticloud FZ LLE

1. Core Hypothesis

Current standard theory calculates Λ_{vac} by summing zero-point energies $((1/2)\hbar\omega)$ up to a cutoff scale (Planck), assuming the vacuum is a static, thermalized ground state. This yields $O(10^{122})$ discrepancy with observation.

Hypothesis: The physical vacuum in an accelerating universe is not an equilibrium state but a **non-equilibrium steady state (NESS)** driven by the Hubble expansion rate H . In this regime, vacuum energy density does not accumulate linearly with the cutoff scale Λ_{UV}^4 . Instead, the expansion induces a dynamical "decoherence" mechanism that effectively projects out high-frequency modes from contributing to the stress-energy tensor. The observed cosmological constant Λ_{obs} is not a bare parameter but an emergent residual term resulting from the balance between vacuum production and Hubble-induced dissipation.

Mathematically, we propose that the effective vacuum energy scales as:

$$\rho_{\text{eff}} \approx \alpha \cdot M_{\text{Pl}}^2 H(t)^2$$

where α is a dimensionless coupling derived from the non-equilibrium entropy production rate, rather than $\rho_{\text{eff}} \sim \Lambda_{\text{UV}}^4$.

2. Mathematical Framework

We construct the framework using **Closed-Time-Path (CTP) Effective Action** formalism extended to curved spacetime with a running renormalization group (RG) flow dependent on the curvature scale.

A. The CTP Generating Functional:

Instead of the standard Euclidean path integral, we utilize the Schwinger-Keldysh contour C to handle time evolution in an expanding Friedmann-Lemaître-Robertson-Walker (FLRW) metric:

$$Z[J] = \int \mathcal{D}\phi_+ \mathcal{D}\phi_- \left[i S[\phi_+] - i S[\phi_-] + i \int d^4x (J_+ \phi_+ - J_- \phi_-) \right]$$

Here, $\phi_{\pm}$ represent fields on the forward and backward branches of the time contour.

B. Curvature-Dependent RG Flow:

We introduce a modified beta function for the vacuum energy density $\rho(\mu)$ where the renormalization scale μ is dynamically tied to the local curvature radius $R^{1/2}$ (or equivalently H):

$$\mu(H) = c_1 H$$

The Callan-Symanzik equation becomes:

$$\left(\mu \frac{\partial}{\partial \mu} + \beta_\lambda \frac{\partial}{\partial \lambda} - 2\gamma \right) \Gamma[\phi; g_{\mu\nu}] = 0$$

Crucially, in the NEQG limit, the counter-term required to cancel the divergence at Λ_{UV} is no longer a constant but a functional of H :

$$\delta \Lambda(H) = - (1/64\pi^2) (\Lambda_{UV}^2 H^2) \cdot f(R, T_{\mu\nu})$$

C. The Dissipative Stress-Energy Tensor:

The expectation value of the stress-energy tensor acquires an imaginary component in the effective action, representing irreversible entropy production due to particle creation in the expanding background:

$$\langle T_{\mu\nu} \rangle_{\text{eff}} = T_{\mu\nu}^{\text{classical}} + \text{Im}(\Gamma_{\text{CTP}})$$

In the infrared limit ($k \rightarrow 0$), the high-frequency modes decouple via a damping factor $e^{-\tau/\tau_H}$ where $\tau_H \sim H^{-1}$. This leads to the cancellation of the quartic divergence:

$$\rho_{\text{vac}}^{\text{ren}} \propto \int_0^\infty dk \sqrt{k^2 + m^2} e^{-k/H} \sim \Lambda_{UV}^4 \mathcal{O}(H^4)$$

3. Proof Sketch / Key Steps

Step 1: Mode Decomposition in FLRW Background

Expand the scalar field $\phi(x,t)$ into conformal time modes η . Show that for a massless minimally coupled field, the mode functions $u_k(\eta)$ exhibit logarithmic growth in the super-horizon limit ($k < aH$).

Result: Standard Bunch-Davies vacuum assumption fails to capture the back-reaction of these growing modes on the geometry.

Step 2: Application of the CTP Formalism

Calculate the one-loop correction to the effective potential $V_{\text{eff}}(\phi)$ using the CTP propagator G_{++} , G_{+-} .

Key Insight: The off-diagonal terms G_{+-} contain information about the initial state's purity. If the universe began in a generic

Conclusion

This framework presents structured pathways toward resolving the Cosmological Constant Problem. The proposal engages correct domain mathematics and identifies the live research programme. Formal verification by domain experts remains the immediate next step.

References

- [1] Witten, E. (1994). Monopoles and four-manifolds. *Math. Res. Lett.*, 1(6), 769–796.
- [2] Penrose, R. (1965). Gravitational collapse and space-time singularities. *PRL*, 14(3), 57–59.
- [3] Hawking, S.W. (1975). Particle creation by black holes. *Commun. Math. Phys.*, 43, 199–220.
- [4] Clay Mathematics Institute (2000). Millennium Prize Problems. CMI Publications.
- [5] Anticloud FZ LLE (2026). PAX L5 Narrow L2 General 27B Technical Report. Internal.
- [6] Alpasan, L.-K. (2026). AI-Assisted Frameworks for Open Problems in Science. Anticloud Research.

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